

1 Define the prediction task

Set the delivery endpoint, held-screen setting, and candidate-ranking decision.

2 Assemble formulation inputs

Map lipid roles, ratios, process, target context, and endpoint support.

3 Control shortcut signals

Quantify source-linked missingness and exclude explicit provenance identifiers.

6 Evaluate reliability

Use screen bootstrap, size, and risk--coverage analyses to define the deployment scope.

5 Train point and rank heads

Optimize masked endpoints and within-screen ranking as separate but coordinated outputs.

4 Split experimental groups

Assign complete groups first; fit preprocessing and select models on training/validation groups.

